

Serial sectioning in the micron-plus range

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The availability of 3-D microstructural information is becoming increasingly important for understanding the complex microstructure-property relationships in advanced materials. Moreover, there is a recent emphasis on the quantification of microstructural features that cannot be reliably obtained via 2-D methods, including: true particle and/or phase morphology and orientation; phase interconnectivity and percolation; particle cracking and damage. Furthermore, many problems in materials science rely on the quantification of the tails of the statistical distributions of these features, with an obvious advantage in going from 2-D microstructural datasets (i.e. micrographs) to 3-D datasets in terms of their dramatically increased volumes of interest.

Mechanical serial sectioning is a well-established technique for obtaining 3-D microstructural data from standard metallographic specimens. Traditionally, layer removal has been done by hand, using standard metallographic polishing techniques such as tripod polishing. This is a very delicate and time-consuming operation: the repeated removal of a fixed depth of material by hand is practically very challenging, and the fidelity of the final 3-D data volume depends entirely on the well-controlled removal of material in each slice.

The tutorial will begin with an introduction to manual serial-sectioning methods, their applicability and limitations. Special emphasis will be given to techniques for maximizing the fidelity of the data, including imaging and alignment issues and some examples of useful image processing tools. Focus will then change to outlining modern methods for automation including robotic specimen manipulation, automatic image capture and automatic specimen prep which can reduce the time to acquire data volumes by around 2 orders of magnitude, compared with manual techniques. Automation will be highlighted as an enabler for systematic 3-D studies of a growing number of different material microstructures, using a variety of experimental set-ups including Spowart and Mullens' fully-automated (robotic) serial sectioning device, **Robo-Met.3D**. The tutorial will also discuss the future of the technique, and show how automated serial sectioning is becoming useful for quality control and failure analysis, in addition to being a unique tool for basic research.

References:

- [1] B. Wunsch, N. Chawla, Serial Sectioning for 3D Visualization and Modeling of SiC Particle Reinforced Aluminum Composites, <http://www.tms.org/Students/Winners/Wunsch/Wunsch.html>
- [2] J. E. Spowart, H. M. Mullens, B. T. Puchala, J.O.M., Vol. 55, No. 10, October 2003.